Backup: Diagnostic 1 component grammars

What each component is, which values it takes, and how many items test it — C1–C5 on SUSHI (CLaSP, ChatTS, TE-3L) and N1–N5 on NOAA (TRACE)

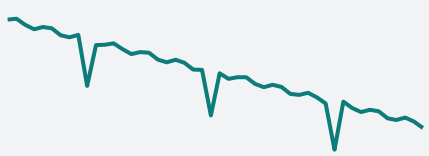


SUSHI grammar

NOAA grammar (N4 planned, could not be built)

One D1 item: the same series, two descriptions

Series (never changed)



illustrative: linear decrease with negative spikes

Correct description

The pattern is witnessing a linear and constant decline. The signal is interrupted by large negative spikes.

Swap item: one component changed (C1)

The increase in the trend is uniform and linear. The signal is interrupted by large negative spikes.

Random control: an unrelated description

a full caption from a class that differs in both label slots

Two two-choice questions per series: correct vs. swap, and correct vs. random. Chance is 0.5 for every component. Gap = random-control accuracy − swap accuracy.

Which grammar each model receives

Model	Data	Components
CLaSP	SUSHI	C1–C5
ChatTS	SUSHI	C1–C5 *
TE-3L (control)	SUSHI	C1–C5
TRACE	NOAA	N1, N2, N3, N5

* ChatTS gets each item as an A/B question, asked in both answer orders.

TRUCE has no D1: its captions are free-form, with no label grammar to swap (left as future work).

SUSHI labels have two slots: fluctuation; shape

SUSHI

Fluctuation slot · 7 values → C4

clean | smooth | noisy | step | positive spike | negative spike | positive-and-negative spike

clean = no fluctuation sentence, so it is never swapped

Shape slot · 20 values → C1, C2, C3, C5

15 trend shapes: constant; linear increase / decrease; exponential growth / decay; inverted exponential growth / decay; sigmoid / inverted sigmoid; gaussian / inverted gaussian; concave / convex; cubic / negative cubic function

5 periodic waveforms: sinusoidal, square, triangle, sawtooth, reverse sawtooth wave

Caption anatomy (series n_spike\02\0000000)

Class label: **negative spike; linear decrease**

SHAPE SENTENCE(S)

"The pattern is witnessing a linear and constant decline."

C1, C2, C3, C5 replace this part

FLUCTUATION SENTENCE (always the last one)

"The signal is interrupted by large negative spikes."

C4 replaces this part

Only series whose shape part is a single sentence are used (1,313 of 1,400), so a swap never changes the sentence count.

7 × 20 = 140 classes · every combination exists · 10 series per class = 1,400 series of 2,048 points

The 20 shapes, by regime, family and direction

SUSHI

Regime	Family	Rising / rises first	Falling / falls first
Trend (15)	**linear**	linear increase	linear decrease
	exponential	exponential growth inverted exponential decay	exponential decay inverted exponential growth
	sigmoid	sigmoid	inverted sigmoid
	Gaussian	gaussian (rise, then fall)	inverted gaussian (fall, then rise)
	quadratic	concave (rise, then fall)	convex (fall, then rise)
	cubic	cubic function	negative cubic function
	constant	*constant — no direction*	
Periodic (5)	**sawtooth**	sawtooth wave (rising ramp)	reverse sawtooth wave (falling ramp)
	other waves	*sinusoidal, square, triangle wave — no direction*	

How the components move in this table

C1 same family, other direction
8 pairs

C2 same direction, other family
16 pairs

C3 one waveform to another
10 pairs

C5 any trend ↔ any waveform
75 pairs

constant has no direction and no family partner, so it appears only in C5. C4 works on the other label slot.

C1–C5 at a glance

SUSHI

Component	The swap changes…	…and keeps	Value pairs	Example (from → to)	Swap items / series
C1 trend direction	the direction of the shape	curve type; fluctuation sentence	8 opposite pairs	linear decrease → linear increase	205 / 205
C2 trend family	the curve type	direction; fluctuation sentence	16 pairs	linear decrease → exponential decay	410 / 177
C3 periodic waveform	the wave shape	periodic regime; fluctuation sentence	10 pairs (all pairs of 5)	sinusoidal → square wave	280 / 70
C4 fluctuation type †	the fluctuation type	the shape sentence	15 pairs (6 types; clean excluded)	negative spike → positive spike	990 generated, 738 certified / 237
C5 signal regime	trend ↔ periodic shape	fluctuation sentence	75 pairs (15 trend × 5 periodic)	linear increase → sawtooth wave	885 / 248

Every pair is used in both directions, and every swap item is paired with a random control on the same series: **2,770 swap + 2,770 random = 5,540 items from 279 of the 280 validation/test series.** † C4 results use the 738 human-certified swaps (thesis Section 3.7, Appendix B.2–B.3).

C1 and C2: which shapes are swapped with which

SUSHI

C1 trend direction · 8 opposite pairs

linear increase	↔	linear decrease
exponential growth	↔	exponential decay
inverted exponential decay	↔	inverted exponential growth
sigmoid	↔	inverted sigmoid
gaussian	↔	inverted gaussian
concave	↔	convex
cubic function	↔	negative cubic function
sawtooth wave	↔	reverse sawtooth wave

The sawtooth pair is the only periodic C1 pair; it is also one of the C3 pairs.

C2 trend family · 16 pairs

Every pair of shapes that share a direction but belong to different families:

Rising 5 pairs

linear increase, exponential growth, inverted exponential decay, sigmoid

Falling 5 pairs

linear decrease, exponential decay, inverted exponential growth, inverted sigmoid

Rises first 3 pairs

concave, gaussian, cubic function

Falls first 3 pairs

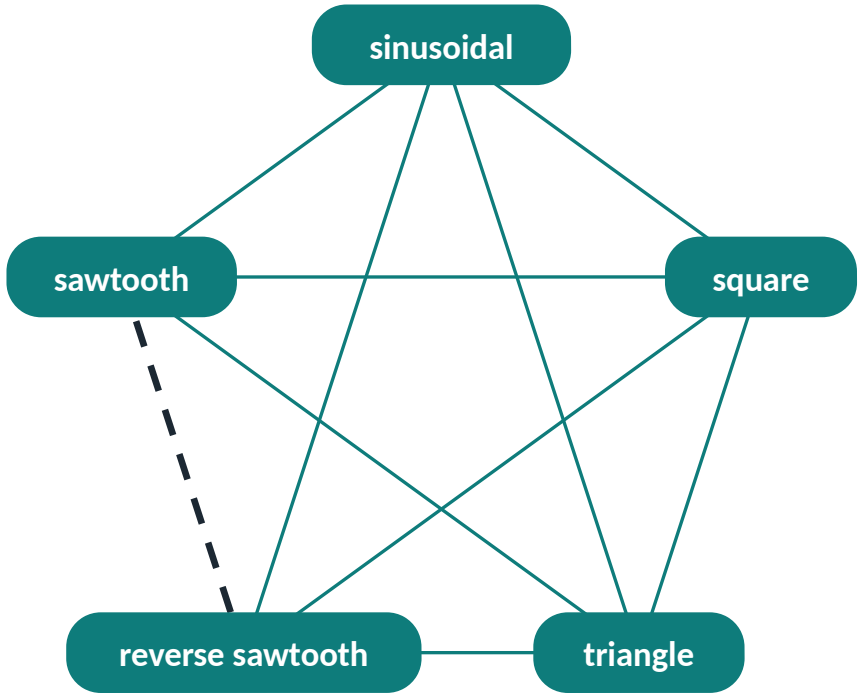
convex, inverted gaussian, negative cubic function

Exponential growth and inverted exponential decay share a family, so they are not a C2 pair (likewise for the falling pair). Constant is in neither C1 nor C2.

C3 and C5: waveform and regime swaps

SUSHI

C3 periodic waveform · all 10 pairs of 5 waves



dashed: sawtooth ↔ reverse sawtooth, also a C1 pair

C5 signal regime · 15 × 5 = 75 pairs

15 trend shapes

constant
linear increase · decrease
exponential growth · decay
inverted exponential growth · decay
sigmoid · inverted sigmoid
gaussian · inverted gaussian
concave · convex
cubic · negative cubic

↔

5 periodic waves

sinusoidal
square
triangle
sawtooth
reverse sawtooth

The largest semantic change in D1: any trend shape against any waveform, both directions (885 swap items, 248 series).

C4 fluctuation type: values and item certification

SUSHI

C4

6 swappable values · 15 pairs

smooth	*"the signal is smooth with minor noise"*
noisy	*"peppered with significant amounts of noise"*
step	*"large step changes"*
positive spike	*"large positive spikes"*
negative spike	*"large negative spikes"*
pos.-and-neg. spike	*"large positive and negative spikes"*
clean	no fluctuation sentence: not swappable (clean ↔ X would add a sentence)

Quotes: typical wording of real SUSHI fluctuation sentences.

From generated to certified swaps



generated

− 107 no explicit type (e.g. "spikes" with no direction)
− 20 word also fits the shape (step changes vs. sawtooth resets)



type stated explicitly → read by hand

− 125 failed the hand check:
subset claim 66 · non-pervasive noise 42 · truncated sentence 10 · reverse overlap 4 · bare noise 3



certified (237 series)

C4 results for CLaSP and ChatTS use the 738; TE-3L uses all 990.

Real SUSHI swaps: one sentence replaced, the rest kept

SUSHI

Comp.	Series class	Original sentence	Replacement sentence
C1	negative spike; linear decrease	“The pattern is witnessing a linear and constant decline.”	“The increase in the trend is uniform and linear.”
C2	pos.-and-neg. spike; linear decrease	“The decrease is consistent in the trend.”	“Sequence aims towards a value by exponential decay.”
C3	smooth; sinusoidal wave	“The signal, akin to a sinusoidal wave, demonstrates a pattern that reoccurs at regular intervals.”	“Comparable to square waves, the signal exhibits a periodic tendency to reverberate at regular intervals.”
C4 †	negative spike; linear decrease	“The signal is interrupted by large negative spikes.”	“Large positive spikes are throughout the signal, additionally.”
C5	noisy; linear increase	“The series observes linear elevation.”	“Signal displays a regular pattern akin to sawtooth waves.”

C1 and C4 edit the two halves of the same caption (series n_spike\02\0000000). Items are from the 50-item validation sample (46/50 valid) and the C4 census; † census-certified.

Why TRACE gets its own grammar on NOAA

NOAA

SUSHI cannot be given to TRACE

2,048 → 186	points: SUSHI series must be shortened to TRACE's maximum input
0.929 → 0.773	feature-classifier accuracy on fluctuation pairs after shortening
0.525–0.595	the spike-related pairs: close to chance
0.93–0.99	global-shape components: unaffected

A failure on shortened SUSHI could not be attributed to TRACE, so TRACE is tested on its own data: NOAA weather, 2,006 test records, 7 variables.

Each NOAA description has ten text fields

Field	Example (record 206)	Swapped by
DATE	The past six months from July 2022 to December 2022.	N2
labels	Clear, Hot, Cold, Cloudy, Humid	N1
location	Mariposa, California experienced a diverse range of …	N5
temperature	… a noticeable trend of rising during the summer months …	N3
visibility	Visibility remained mostly stable, averaging around 16.09 km …	*(N4, dropped)*
5 other fields	precipitation, relative humidity, wind u/v, sky cover	never edited

A swap edits one field; the other nine fields and all seven series stay unchanged, and the fields are joined into TRACE's description template.

N1–N5 at a glance

NOAA

Component	The swap changes…	Values	Eligible → swap items	Example (from → to)
N1 weather labels	every label that has an opposite, all at once	Hot ↔ Cold, Warm ↔ Cool, Clear ↔ Cloudy, Rainy ↔ Dry; other labels kept	462 → 400	Clear, Warm, Dry → Cloudy, Cool, Rainy
N2 time span	the stated span; end date kept, start date recomputed	one week ↔ six months (28-day records not used)	1,256 → 400	the past week → the past six months
N3 temperature trend †	rising / warming words ↔ falling / cooling words; stated values unchanged	up ↔ down	777 → 400 → 389 → 344 certified	upward trend → downward trend
N4 fluctuation vs. stability	*fluctuating ↔ stable wording*	*carrier fields: temperature, visibility*	*33 records (< 100 pre-set minimum)*	*dropped, not run*
N5 location	the location sentence, taken from a record in another US state	US places	2,006 → 400	Polk, Iowa → Ocean, New Jersey

400 swap + 400 random-control items per component (seed 42); the random control is the full description of an unrelated record. **N5 was meant to be undetectable from the series, yet TRACE scores 0.918–0.935 on it.** † Only N3 was checked item by item.

Real NOAA swaps and the rules that filter records

NOAA

Examples from the item-validation sheets

N1 — labels

Clear, Warm, Dry
→ **Cloudy, Cool, Rainy**

N2 — DATE

The past week spans from June 28, 2020, to July 4, 2020.
→ **The past six months from January 4, 2020 to July 4, 2020.**

N3 † — temperature

… ranging from a low of −4.33°C to a high of 17.0°C, with a noticeable upward trend towards the end of the period.
→ **… with a noticeable downward trend towards the end of the period.**

N5 — location

Smith, Texas experienced a variety of weather conditions.
→ **The weather report is for Vermilion Parish, Louisiana.**

† certified by the full N3 hand check. N1, N2 and N5 items received automated checks only.

Records excluded before sampling

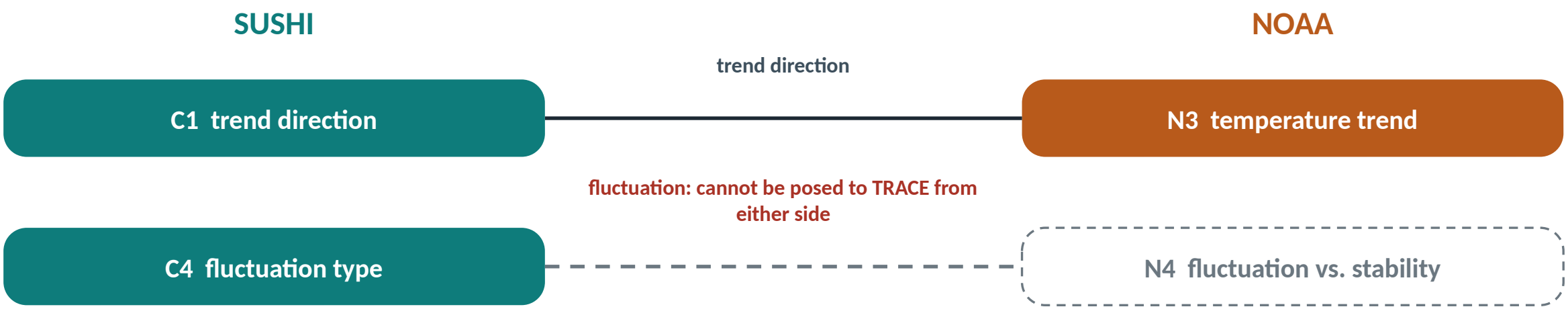
N1 label set already contradictory (e.g. Hot + Cold over six months): 1,345; contradictory after the swap (e.g. Dry + Humid): 199

N2 28-day spans, to keep the swap binary: 679; dates that could not be parsed: 71

N3 both directions in the field 382 · months, seasons or times of day that fix the direction 606 · numbers that reveal it (“starting from …”) 145 · no trend word 93 · trend tied to a value 3; after generation, 11 + 45 removed by checks

N4 fluctuation claims almost always cite values or ranges a one-word edit cannot keep consistent: only 33 records left

How the two grammars line up



Only in SUSHI

C2 trend family
C3 periodic waveform
C5 signal regime

Numbers are compared within a model only; across models the thesis compares relative degradation and verdicts (Table 4.3).

Only in NOAA

N1 weather labels
N2 time span
N5 location

Fluctuation: SUSHI loses spike information when shortened for TRACE, and NOAA admits only 33 consistent N4 edits.